

Space Shooter - Complete Build Instructions for Windows

Prerequisites

1. **Unity Hub** - Download from: <https://unity.com/download>
 2. **Unity Editor** - Version **2021.3 LTS** or newer (2022.3 LTS recommended)
 - During install, ensure **Windows Build Support (IL2CPP)** module is included
 - Also install **Windows Build Support (Mono)** as a fallback
 3. **Windows 10/11** (64-bit) for building
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Step-by-Step Setup

Step 1: Create a New Unity Project

1. Open **Unity Hub**
2. Click **"New Project"**
3. Select the **"2D (Built-in Render Pipeline)"** template
4. Name it `SpaceShooter`
5. Choose your preferred location
6. Click **"Create Project"**

Step 2: Copy Project Files

1. Close Unity (or keep it open - it will auto-refresh)
2. Copy the contents of this project into your Unity project:

```
# Copy all script files:  
Copy the entire Assets/ folder contents INTO your Unity project's Assets/ folder:  
- Assets/Scripts/           → YourProject/Assets/Scripts/  
- Assets/Editor/            → YourProject/Assets/Editor/
```

If Unity is open, it will automatically detect and import the new files.

Alternatively, you can replace the entire Assets folder:

```
# Delete the existing Assets folder in your Unity project  
# Copy our Assets folder into the Unity project root
```

Step 3: Set Up the Game Scene (One-Click Method)

1. Open Unity and wait for scripts to compile (check bottom-right for progress)
2. In the menu bar, go to: **Tools → Space Shooter → Create New Scene and Setup**
3. This automatically:
 - Creates a new scene
 - Sets up the camera (orthographic, dark space background)

- Adds the GameBootstrapper (which creates everything else at runtime)
- Saves the scene to `Assets/Scenes/GameScene.unity`

Step 4: Test in Editor

1. Press the **Play** button (▶) in the Unity Editor
2. You should see:
 - **Main Menu** with “SPACE SHOOTER” title
 - Click “**START GAME**” to begin playing
 - Stars scrolling in the background
 - Your blue player ship at the bottom
3. Controls:
 - **WASD** or **Arrow Keys**: Move
 - **Space**: Shoot
 - **ESC**: Pause

Step 5: Configure Build Settings

Automatic Method:

1. Go to: **Tools → Space Shooter → Configure Build Settings**
2. This sets resolution, window mode, and build target automatically

Manual Method:

1. Go to **File → Build Settings**
2. Click “**Add Open Scenes**” to add GameScene
3. Select “**PC, Mac & Linux Standalone**” as platform
4. Click “**Switch Platform**” if needed
5. Under **Target Platform**, select “**Windows**”
6. Set **Architecture** to “**x86_64**”

Player Settings:

1. Click “**Player Settings...**” in Build Settings window
2. Configure:
 - **Company Name**: SpaceShooterDev (or your name)
 - **Product Name**: Space Shooter
 - **Default Screen Width**: 1920
 - **Default Screen Height**: 1080
 - **Fullscreen Mode**: Fullscreen Window
 - **Resizable Window**: ✓ (checked)
 - **Run In Background**: ✗ (unchecked)

Step 6: Build the Game

Quick Build Method:

1. Go to: **Tools → Space Shooter → Build Windows Executable**
2. Choose an output folder (e.g., create a `Builds` folder)
3. Wait for the build to complete
4. The executable will be at: `Builds/SpaceShooter.exe`

Manual Build Method:

1. Go to **File → Build Settings**
2. Click “**Build**”

3. Choose output folder
4. Name the executable `SpaceShooter.exe`
5. Click **“Save”**
6. Wait for the build process to complete

Step 7: Run the Game

1. Navigate to your build output folder
2. Double-click `SpaceShooter.exe`
3. The game launches in fullscreen mode
4. Enjoy!

Build Output Structure

After building, your output folder will contain:

```
Builds/
├── SpaceShooter.exe           # The game executable
├── SpaceShooter_Data/        # Game data folder (required)
│   ├── Managed/              # .NET assemblies
│   ├── Resources/            # Game resources
│   ├── level0                # Scene data
│   └── ...
├── MonoBleedingEdge/         # Mono runtime (if Mono build)
└── UnityCrashHandler64.exe    # Crash handler
```

Important: To distribute the game, you must include the ENTIRE build folder, not just the .exe file.

Distribution

To share your game:

1. Zip the entire build output folder
2. The recipient just needs to extract and run `SpaceShooter.exe`
3. No Unity installation required for players
4. Works on Windows 10/11 (64-bit)

Troubleshooting

“Scripts have compiler errors”

- Ensure you’re using Unity 2021.3+ or 2022.3+
- Check that ALL script files are in the correct folders
- Go to **Edit → Preferences → External Tools** and click **Regenerate Project Files**

“No scenes in build”

- Open your GameScene
- Go to **File → Build Settings**
- Click **“Add Open Scenes”**

“Build fails with IL2CPP errors”

- Try switching to Mono scripting backend:
- **Edit → Project Settings → Player → Other Settings**
- Change **Scripting Backend** from IL2CPP to Mono

Game runs but no objects appear

- Make sure the GameBootstrapper object exists in the scene
- Check the Console (Window → Console) for error messages
- Verify all scripts compiled without errors

Font not displaying

- The game uses Unity’s built-in Arial font
- If text appears as boxes, the built-in font may not be available
- This is rare but can happen in very old Unity versions

Optional: Adding Sound Effects

The game has full audio integration but ships without audio files.
To add sounds:

1. Download free SFX from sites like:
 - <https://freesound.org>
 - <https://opengameart.org>
 - <https://kenney.nl/assets> (free game assets)
2. Import `.wav` or `.ogg` files into `Assets/Audio/`
3. Select the **AudioManager** GameObject in the scene
(or find it in the Hierarchy at runtime)
4. Drag audio clips to the corresponding slots:
 - Player Shoot Clip
 - Enemy Shoot Clip
 - Player Hit Clip
 - Enemy Explosion Clip
 - Player Explosion Clip
 - Power Up Clip
 - Shield Break Clip
 - Game Over Clip
 - Background Music

The game works perfectly without sounds - the AudioManager gracefully handles null/missing clips.

System Requirements (for the built game)

- **OS:** Windows 10/11 (64-bit)
- **RAM:** 512 MB
- **GPU:** Any GPU with DirectX 11 support
- **Storage:** ~100 MB
- **Input:** Keyboard